ACS APPLIED MATERIALS & INTERFACES

Research Article

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Solution-Processed Organic and Halide Perovskite Transistors on Hydrophobic Surfaces

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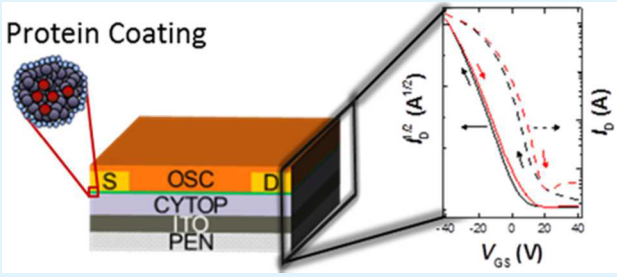
ABSTRACT: Solution-processable electronic devices are highly desirable due to their low cost and compatibility with flexible substrates. However, they are often challenging to fabricate due to the hydrophobic nature of the surfaces of the constituent layers. Here, we use a protein solution to modify the surface properties and to improve the wettability of the fluoropolymer dielectric Cytop. The engineered hydrophilic surface is successfully incorporated in bottom-gate solution-deposited organic field-effect transistors (OFETs) and hybrid organic−inorganic trihalide perovskite field-effect transistors (HTP-FETs) fabricated on flexible substrates. Our analysis of the density of trapping states at the semiconductor−dielectric interface suggests that the increase in the trap density as a result of the chemical treatment is minimal. As a result, the devices exhibit good charge carrier mobilities, near-zero threshold voltages, and low electrical hysteresis.



KEYWORDS: *organic thin-film transistors, flexible electronics, organic semiconductors, halide perovskites, charge carrier mobility*

1. INTRODUCTION

The fabrication of electronic and optoelectronic devices using solution-based techniques offers compelling advantages, including ease of manufacturing, low cost, and excellent compliance with high-throughput processing.[1,2] Significant research effort was channeled toward the development of solution-processable electronic materials, resulting in a steadily increasing number of available soluble semiconductor, electrode, and dielectric materials.[2−9] These compounds were incorporated into various types of devices such as photovoltaic cells, light-emitting diodes, transistors, memristors, or sensors. Most of these devices, however, had only one or a few layers deposited from solution, using simple methods such as inkjet printing, spin-casting, or spray coating, while the other layers were obtained by electron-beam evaporation, thermal evaporation, or other complex technologies, which significantly increased the overall cost of fabrication and limited their applicability. This compromise was necessary, however, because the hydrophobic nature of many solution-deposited compounds precluded the deposition of consecutive layers onto their surface. To expand upon the use of such materials, devices were designed such that the hydrophobic layer was deposited in the last stages of device fabrication. While these design methodologies have broadened the applicability of many newly developed compounds, their compatibility with a limited set of device architectures restricts the applications that they can address. For example, the fluoropolymer dielectric Cytop was incorporated into high-mobility organic field-effect transistors (OFETs) exhibiting minimal bias-stress effects because its inert nature yielded a low density of electronic trap states at its interface with the semiconductors.[10−16] This property also allowed the fabrication of the first organic−inorganic hybrid trihalide perovskite field-effect transistor (HTP-FET) that operated at room temperature.[17] While in that report the mobility values and device yields were modest, more recent work that explored the Cytop/perovskite interface reported impressive transistor characteristics.[18] In all of these studies, however, this dielectric could only be incorporated in devices with top-gate geometry, having bottom contacts (Figure 1a) or top contacts (Figure 1b). FETs using this dielectric in a bottom-gate structure (Figure 1c,d) have not been possible because the low surface energy of the Cytop layer prevented the adhesion of any solution-deposited semiconductor layers on its surface.

In this article we present a method for tailoring the surface properties of the highly hydrophobic dielectric Cytop by applying a treatment with a protein solution to improve its wettability and allow for subsequent solution processing at its surface. We find that this treatment does not alter the electrical properties of the Cytop layer, but increases its surface energy sufficiently to allow for solution deposition of other layers on its

Received: March 6, 2017
Accepted: May 9, 2017
Published: May 9, 2017

ACS Publications © 2017 American Chemical Society

DOI: 10.1021/acsami.7b03232
ACS Appl. Mater. Interfaces 2017, 9, 18120−18126

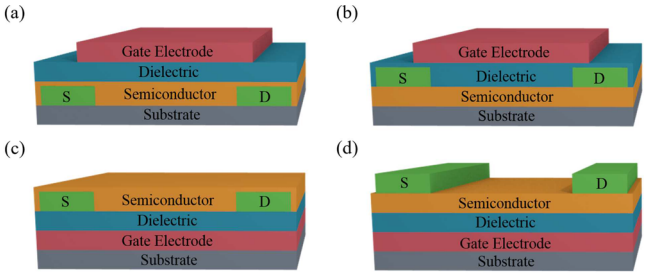


Figure 1. Schematic diagrams of the four possible architectures of field-effect transistors, where S denotes the source electrode and D denotes the drain electrode: (a) top-gate bottom contact, (b) top-gate top contact, (c) bottom-gate bottom contact, and (d) bottom-gate top contact.

surface. We then fabricate bottom-gate FET devices (Figure 1c,d) on the protein/Cytop dielectric with both organic and halide perovskite semiconductors. These two classes of materials have been chosen precisely because of their compatibility with solution-processing methods, which makes them serious contenders for low-cost electronic devices. We obtained transistor device performance similar to or better than devices made on silicon dioxide (SiO_2), the most commonly used dielectric in bottom-gate FET devices. Here we achieve this performance with the additional advantages of solution processability and compatibility with flexible and transparent substrates. The dielectric surface modification proposed in this study is different than the one in which silane-based self-assembled monolayers (SAMs) are used. Unlike the SAM treatment,[19−21] which results in the formation of a covalent bond between the monolayer and the oxide dielectric, our technique takes advantage of the tendency of proteins to spontaneously adsorb to hydrophobic surfaces.[22−24] We have chosen to use a protein solution extracted from albumen, as it is widely available and easy to prepare.

2. EXPERIMENTAL DETAILS

2.1. Dielectric Fabrication and Characterization. Flexible substrates consisting of polyethylene terephthalate (PET) coated with 130 nm indium tin oxide (ITO) (639303-5EA, Sigma-Aldrich, St. Louis) were first heated to a temperature of 120 °C for 30 min, followed by a slow cooling process. This preshrinking step was necessary to ensure that the film remained flat and unbuckled during the heating steps necessary for FET fabrication. The substrate was cleaned with hot (85 °C) isopropyl alcohol and dried with a nitrogen stream. The final cleaning step was a 10 min exposure to UV ozone, a 30 s rinse with deionized (DI) water, and a thorough drying with a nitrogen stream. The Cytop layer was then deposited by spin-casting the fluoropolymer Cytop (Asahi Glass, distributed by Bellerex Inc., type 809-A) over the ITO/PET substrate at a spinning speed of 2000 rpm for 60 s. The polymer film was first cured at 50 °C for 30 min on a hot plate in a glovebox (<0.1 ppm of H_2O, <0.1 ppm of O_2) and then heated at 110 °C for at least 1 h under vacuum.

The thickness of the Cytop dielectric film was evaluated by atomic force microscopy (AFM) using a Topometrix Explorer AFM (Veeco Instruments, USA). The sample was glued to a metallic substrate and placed onto a movable x-y stage. The internal camera of the AFM allowed for visualization of the sample's position. The thickness was evaluated before and after the protein treatment using a rectangular silicon cantilever in tapping mode (Nanosensors, spring constant, k = 21−98 N/m; resonance frequency, f = 146−236 kHz). Large areas (100 × 100 μm) were first scanned using a fast rate of 200 μm/s to determine the position of the cantilever tip relative to the Cytop and ITO surfaces. Once the step location was found, a slower rate of 25 μm/s over a 50 × 50 μm area was used to measure the thickness with a higher resolution. The roughness and the thickness of the protein layer were determined by AFM using an Asylum MFP-3D-Bio AFM (Asylum Research, USA) at ambient conditions. A rectangular silicon cantilever in tapping mode (Mikromasch, spring constant, k = 25−60 N/m; resonance frequency, f = 150−190 kHz) was used. Images (1024 × 1024 pixels) were typically collected at a scan rate of 1−3 lines per second. Approximately 1 × 1 cm samples of protein-treated Cytop were attached with superglue to a glass coverslip and placed in the AFM. Multiple scans of 50, 5, and 1 μm were collected to confirm uniformity of the samples. The images were flattened with a second-order (or lower order) polynomial to remove large-scale bowing or slant. Roughness of the samples was determined from 1 μm scans. To determine the thickness of the protein layer, the protein-treated Cytop sample was scratched with 30 nN of normal force. This level of force removed the protein layer from the Cytop surface, but it did not leave a scratch in the Cytop (as tested on the plain Cytop surface).

2.2. Surface Treatment. The protein solutions were prepared by first extracting the liquid albumen (whites) from a chicken egg and then dissolving the protein-rich liquid in DI water. The weight concentration of protein contained within the solution was determined using the mass of liquid albumen and DI water. The solution contains mostly ovalbumin, but other proteins like ovotransferrin, ovomucoid, ovomucin, lysozyme, and avidin are also present.[25] The surface of the Cytop layer was modified by either placing a few drops of the protein solution onto the hydrophobic surface or by immersing the PET/ITO/Cytop substrate in the solution facing upside down for 10 min, followed by a thorough rinsing with DI water and drying using a nitrogen stream. The second technique produced better quality films, so we used it for all device work. The surface energies of the untreated and of the protein-modified Cytop were characterized using the static sessile drop method with DI water. The water contact angles were measured using a Ramé-Hart Model 200 Contact Angle Goniometer and calculated using DROPimage Standard.[26] The measurements were conducted in air and at room temperature.

2.3. Field-Effect Transistor Fabrication and Characterization. The FETs with modified dielectric surfaces were tested in combination with an organic semiconductor, 2,8-difluoro-5,11-bis-(triethylsilylethynyl) anthradithiophene (diF-TES ADT) (OFETs), and with an organic−inorganic hybrid halide perovskite, $CH_3NH_3PbI_{3-x}Cl_x$ (HTP-FETs), as the semiconductor layer. We fabricated devices of the structure presented in Figure 1c,d with ITO bottom-gate electrode and protein/Cytop dielectric. For the OFET devices, the structure in Figure 1c was adopted. Here the gold source and drain contacts (60 nm) were deposited by thermal evaporation through a shadow mask and then treated with pentafluorobenzenethiol SAM (PFBT, Sigma-Aldrich) by submersing the entire substrate into a 30 mM bath of PFBT dissolved in ethanol (Sigma-Aldrich) for 30 min. SAM deposition was finalized with a thorough rinse with ethanol to ensure complete removal of the bulk layers. This step was necessary to improve charge injection and film morphology.[27] The organic semiconductor was spin-coated over the PFBT/Au contacts at 1000 rpm using a ramping time of 10 s. Several OFET devices of the same structure were fabricated on SiO_2 for reference by following the procedure described elsewhere.[27,28]

We used similar PET/ITO/protein/Cytop substrates for the HTP-FETs. The perovskite layer was fabricated from a solution of 82 mg of CH_3NH_3I, 23.5 mg of PbI_2, and 40 mg of $PbCl_2$ dissolved in 250 μL of *N*,*N*-dimethylformamide, stirred for 10 h at 70 °C under nitrogen, and spin-coated at 3000 rpm over the treated Cytop substrate. Without exposure to air, the obtained film was annealed on a hot plate in a nitrogen-filled glovebox at 105 °C for 2 h. Top source and drain contacts were deposited using thermal evaporation at a rate of 1 Å/s, which resulted in FETs of the structure shown in Figure 1d.

All devices were placed in slight vacuum for 24 h to ensure full solvent removal and were electrically characterized in air and dark using an Agilent 4155C Semiconductor parameter analyzer. Devices of the same geometry, but with an untreated Cytop dielectric layer, were also fabricated for reference. The semiconductor film, however, did not form a continuous film on these surfaces and we were unable to obtain working devices. Cross-polarized optical micrographs of the diF-TES

DOI: 10.1021/acsami.7b03232
ACS Appl. Mater. Interfaces 2017, 9, 18120−18126

ADT thin films that were spin-cast onto neat Cytop and protein-treated Cytop are shown in part (a) and (b), respectively, of Figure 2.

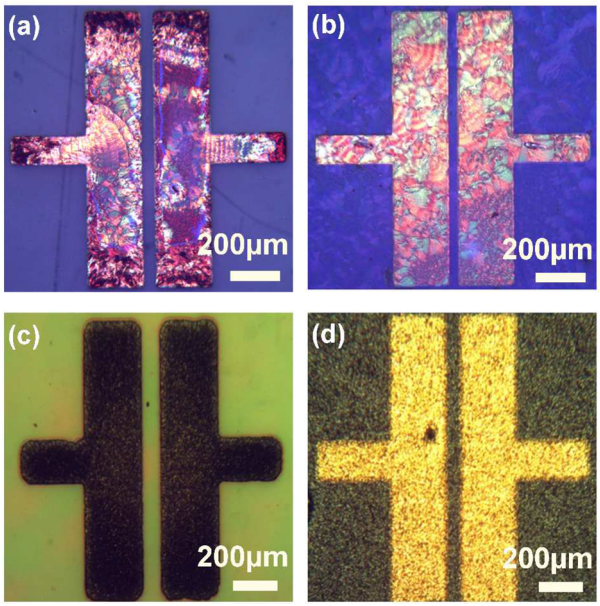


Figure 2. Optical micrographs of the diF-TES ADT thin-film morphology when the solution was spin-cast onto (a) untreated and (b) protein-treated Cytop dielectric. Similarly, a perovskite film spin-cast on to (c) untreated and (d) protein-treated Cytop dielectric is shown. The left and right rectangles in each figure are the source and drain electrodes with the exposed dielectric between.

Similar images were acquired for the perovskite films, which are shown in Figure 2c,d. These images show that due to the hydrophobic nature of the dielectric, the crystalline growth of the semiconductor films deposited on untreated Cytop was isolated to the gold electrode regions where the surface energy is higher.[27] In contrast, the film deposited on the protein-treated Cytop is continuous and homogeneous. This creates an environment suitable for the fabrication of a functioning FET, as we will describe in detail later. Since we were unable to obtain working bottom-gate devices on a neat Cytop dielectric film, we fabricated bottom-contact, top-gate FETs (Figure 1a) with Cytop dielectric and Au top-gate electrode to access the Cytop/semiconductor interface and compare its properties with the Cytop-protein/semiconductor interface.

■ RESULTS AND DISCUSSION

In Figure 3a we show the chemical formula of the Cytop molecule, and in Figure 3b the schematic representation of protein adsorption on a hydrophobic surface, such as that created by the Cytop film. The detailed interactions of protein adsorption onto surfaces are very complex and have not yet been fully resolved. However, guiding principles and rules have emerged over the last 2 decades, mainly from the medical devices field.[23] Proteins that exist in an aqueous environment have mostly hydrophilic amino acids on the water-exposed outside and hydrophobic amino acids on the inside (the hydrophobic core). Most proteins strongly adsorb to hydrophobic surfaces.[23] When they adsorb, it is thought that they partially denature so that the hydrophobic core interacts with the hydrophobic surface, while the hydrophilic amino acids point away from the surface toward the aqueous solution.[23,24]

The contact angle measurement obtained at the surface of untreated Cytop is included in Figure 3c and that for the protein/Cytop surface is shown in Figure 3d. The treatment of

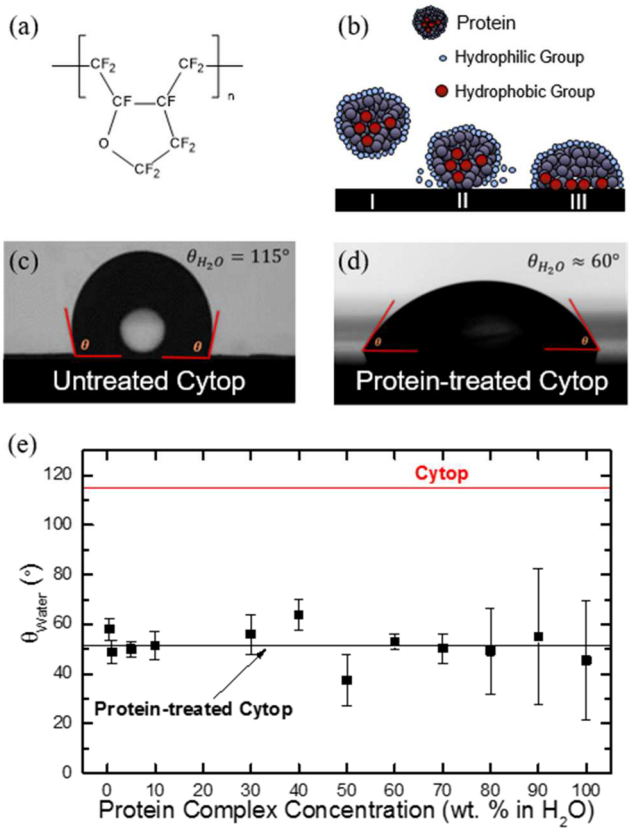


Figure 3. (a) Chemical structure of Cytop. (b) Interaction of protein molecules with hydrophobic substrates. (c) Optical image of a water droplet on (c) untreated and (d) protein-treated Cytop. (e) The water contact angle (θ_{Water}) as a function of the protein solution concentration used to treat the Cytop surface. The red line denotes the water contact angle of neat Cytop.

the Cytop film with the aqueous protein solution produced a decrease in the water contact angle (θ_{Water}) from $\theta_{Water} = 115°$, characteristic of a Cytop surface, to $\theta_{Water} \approx 60°$. We found that the contact angle is independent of the protein solution concentration (see Figure 3e) and a hydrophilic surface is obtained even for protein concentrations as low as 0.5 wt % in DI water. This is likely due to the fact that even in the dilute solutions, the protein content is sufficient to form one monolayer at the surface of Cytop. Any additional adsorbed layers will be removed during the rinsing step due to the weak affinity among the protein molecules. We therefore used a concentration of 1% for all our device studies.

Figure 4 shows the AFM measurements of the step height obtained on the neat Cytop film (Figure 4a) and the protein-treated Cytop (Figure 4b), while in Figure 4c we include the surface roughness measurement at the surface of the protein-treated Cytop. The thickness of the dielectric was found to be d = 750 nm, with a negligible change after treatment; the protein layer thickness was estimated to be 2 nm (see Figure 4d), while the roughness was determined to be 0.3 nm. The thickness, together with the dielectric constant of Cytop, ε_r = 2.05, was used to determine the areal capacitance of the dielectric film, C_i, which was found to be C_i = 2.42 nF·cm^{-2}. This value was confirmed by our capacitance measurements using the ITO/Cytop/protein/Au geometry.

The diF-TES ADT FETs fabricated on bottom-gate protein-treated Cytop dielectric film were measured at room temper-

DOI: 10.1021/acsami.7b03232
ACS Appl. Mater. Interfaces 2017, 9, 18120−18126

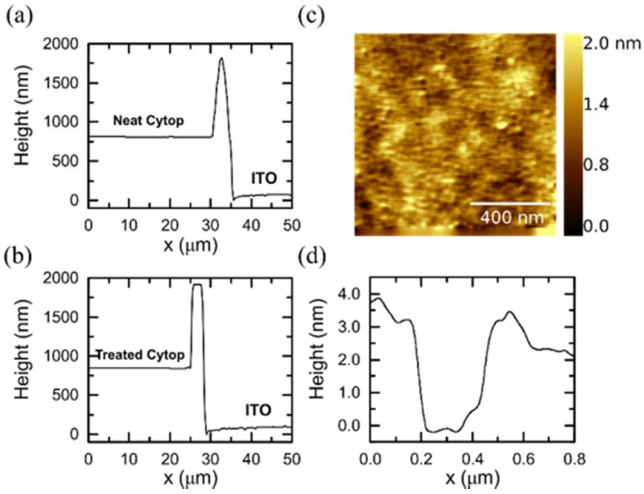


Figure 4. Step-height plot demonstrating the measured thickness of (a) Cytop and (b) protein-treated Cytop on the ITO/PET surface. (c) AFM topography image of protein-treated Cytop and (d) step-height profile used for thickness measurement of the protein layer.

ature and representative current−voltage characteristics are shown in Figure 5a,b. The evolution of the drain current (I_D) as a function of the gate-source voltage (V_{GS}) in the saturation regime of device operation is shown in Figure 5a, which includes the device geometry as inset. Here, the drain-source voltage (V_{DS}) is −40 V. The small hysteresis suggests that the addition of protein molecules at the semiconductor/dielectric interface, where the transistor channel forms, does not increase significantly the density of traps. A more thorough discussion on this topic will be provided later. We determined the field-effect mobility using

$$\mu_{sat} = \frac{2L}{WC_i}S^2 \quad (1)$$

where S is the slope of the $\sqrt{I_{DS}}$ versus V_{GS} plot (left axis in Figure 5a) and L and W are the channel length and width, respectively.[2,3] The device represented in Figure 5a yielded a field-effect mobility of μ = 0.26 cm^2 V^{-1} s^{-1}.

Figure 5b shows the evolution of the drain current with the drain voltage for a constant gate-source voltage that was varied between 0 and −50 V in steps of −10 V. A p-type transport is observed, with some deviations from linearity in the low-voltage regime, indicating that the contact resistance is not negligible.[27] The chemical formula of the organic semiconductor is included in the inset. The field-effect mobility and threshold voltage values determined on over 50 devices are summarized in the histograms in Figure 5c,d, respectively. The average mobility was μ = 0.11 ± 0.09 cm^2 V^{-1} s^{-1} and the threshold voltage ranged from −10 to 10 V. These device parameters are similar or superior to those obtained on the same organic semiconductor when it was incorporated in FETs of a similar geometry, but with a SiO_2 dielectric.[27−30] The FETs developed in this study, however, are fabricated on a flexible substrate and the dielectric layer was deposited from solution, with the maximum substrate temperature reached during processing being 110 °C. Nevertheless, we note that the device properties are inferior to those obtained on diF-TES ADT devices with top-gate structure, where an average of 1.5 cm^2 V^{-1} s^{-1}, with a maximum of 4.8 cm^2 V^{-1} s^{-1} was achieved, together with a close-to-zero threshold voltage.[29] Similarly, top-gate FETs fabricated from polymer/diF-TES ADT blends with Cytop dielectric have achieved mobilities greater than 5 cm^2 V^{-1} s^{-1}.[10,11,15] This is partially a result of the differences in the device architecture, as we have shown in our earlier work, but may also originate from the different contribution of trapping states at the interface between the semiconductor and dielectric.[29] To quantitatively access this information, we performed density of states analysis (DOS) on our devices.

The trap density of states was extracted from room temperature I_D −V_{GS} measurements, using the Grünewald's method.[29,31,32] The gate-voltage-dependent surface potential and conductivity provide a sensitive probe into the bandgap states as a function of energy with respect to the Fermi level. In Figure 6 we show the DOS estimated at the Cytop/protein/diF-TES ADT interface in a bottom-gate configuration (blue) and Cytop/diF-TES ADT interface in a top-gate configuration (red). The mobility obtained in this FET with a top Cytop dielectric is μ = 1.1 cm^2 V^{-1} s^{-1}, which is typical for this semiconductor in this device geometry.[29] The two blue curves correspond to Cytop/protein/diF-TES ADT samples of different quality: the data obtained on the high-mobility device (μ = 0.13 cm^2 V^{-1} s^{-1}) is shown in light blue, while that recorded on the low-mobility transistor (μ = 0.05 cm^2 V^{-1} s^{-1}) is shown in dark blue. The inverse dependence of the charge carrier mobility on the density of traps for the devices of identical structure suggests that better control of the interface between the protein/Cytop and the organic semiconductor may consistently lead to higher performing devices. Note that in all FETs with treated dielectric the DOS was greater than in the case of neat Cytop, which is to be expected since this

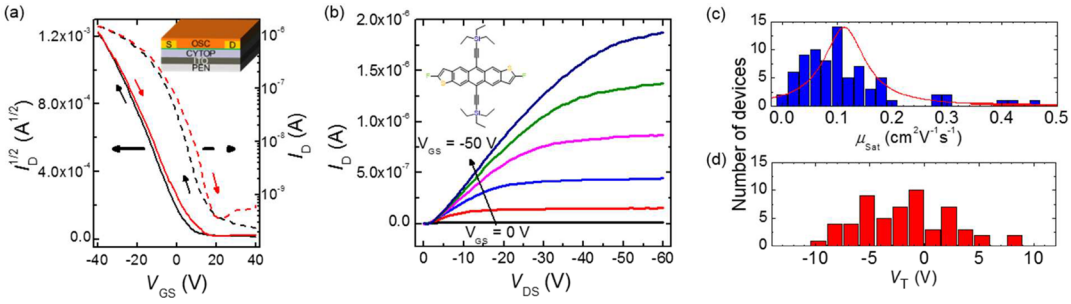


Figure 5. (a) Evolution of the drain current versus the gate-source voltage in the saturation regime for the forward (black) and reverse (red) sweep. The inset shows the device structure, with the protein layer in green. (b) Drain current versus source-drain voltage at constant gate voltage. The chemical structure of diF-TES ADT is shown in the inset. Summary of the transistor characteristics of ∼50 spin-coated devices: (c) saturation mobility (blue); (d) threshold voltage (red).

DOI: 10.1021/acsami.7b03232
ACS Appl. Mater. Interfaces 2017, 9, 18120−18126

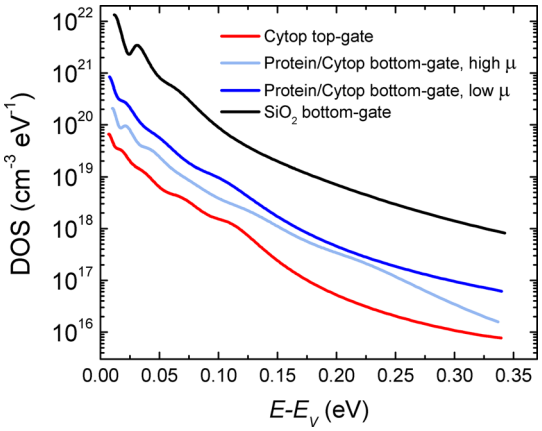


Figure 6. Trap DOS in diF-TES ADT OFETs with different gate dielectrics, as follows: SiO_2 bottom gate (black), protein-treated Cytop bottom gate (blue), and Cytop top gate (red). The dark blue curve corresponds to the DOS measured in a FET of low mobility, while the light blue was obtained in a high mobility device of the same structure.

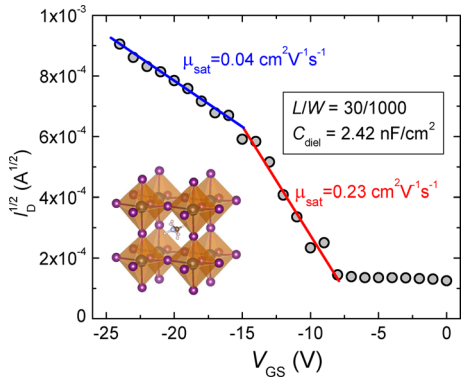


Figure 7. Drain current versus the applied gate-source voltage in the saturation regime, at room temperature, for a FET with protein/Cytop bottom gate. The device geometry and dielectric capacitance, as well as the perovskite structure, are included in the inset.

interface is more hydrophilic. Nevertheless, the density of traps is lower than that obtained at the surface with SiO_2 dielectric (black), in agreement with the lower mobility reached in these FETs (μ = 0.02 cm^2 V^{-1} s^{-1}).

To test how generally applicable the proposed method is, we further applied it for the fabrication of bottom-gate FETs with halide perovskites as the semiconductor layers. This class of materials has gained a lot of attention recently because of its remarkable optical and electrical properties and its solution processability, which are key properties necessary for low-cost optoelectronic applications. Halide perovskites have been incorporated in high-efficiency solar cells and light-emitting diodes,[33,34] tunable lasers,[35] photodetectors,[36] and magnetic devices.[37] The room-temperature operation of trihalide perovskite FETs, however, is limited to a device architecture with Cytop top-gate dielectric (Figure 1a,b) since this dielectric yields a low density of interfacial traps.[17,18,38] The bottom-gate devices (Figure 1c,d) have been reported only at low temperature, with no electrostatic gating observed at temperatures above 250 K.[39,40] The room-temperature operation of bottom-gate trihalide perovskite FETs would expand their use toward new low-cost, thin-film electronic circuits that could potentially lead to integrated optoelectronic systems. In this work, we achieved room-temperature operation in bottom-gate halide perovskite FETs with the structure presented in Figure 1d having an ITO gate electrode, a protein-treated Cytop dielectric, $CH_3NH_3PbI_{3-x}Cl_x$ semiconductor layer, and Au source-drain electrodes. A typical transfer curve is shown in Figure 7. The inset in this figure represents the crystal structure of the perovskite, with the Pb^{2+} cations and the halide anions sitting at the centers and corners of the octahedra, respectively; and the methylammonium organic cations in the interstices. The current−voltage characteristics shown here deviate from a straight line necessary to use eq 1 for mobility estimation. This type of behavior results from severe contact effects, which we have discussed in detail elsewhere.[41] In the low-voltage regime we obtained μ = 0.23 cm^2 V^{-1} s^{-1}, a value which is most likely an overestimation of the real mobility of this device, while the value obtained at high voltages, μ = 0.04 cm^2 V^{-1} s^{-1} is a more conservative estimation. Both values, however, are lower than the mobility obtained on the same perovskite layer, but with Cytop top dielectric, μ = 1.3 cm^2 V^{-1} s^{-1}.[17] Similar to the case of OFETs, this is probably due to increased density of trapping states at the semiconductor/dielectric interface because of the insertion of the protein layer. In addition, while for the top-gate devices with neat Cytop dielectric layer we observed ambipolar transport, we found that the electron transport was inhibited at the hydrophilic interface created by the protein treatment, similar to earlier observations in OFETs.[20]

Our results demonstrate that protein treatment of highly hydrophobic dielectrics can increase their surface energy sufficiently to allow solution processing, while preserving the quality of the interface. While in this work we focused on a protein mixture, further systematic studies of the protein/dielectric and protein/semiconductor interactions for different purified proteins and various semiconductors will elucidate the nature of these interactions. This will allow one to exploit the protein treatment beyond tuning the surface energy, to enhance the crystallinity of the semiconductor film, and fine-tune FET properties such as mobility, threshold voltage, and subthreshold swing.

■ CONCLUSIONS

In summary, we modified the surface energy of Cytop, a highly hydrophobic dielectric layer, by treating it with a protein solution. We showed that this process does not alter the electrical properties of the dielectric, but it makes it more hydrophilic, allowing subsequent solution processing of other layers at its surface. We fabricated bottom-gate FETs with organic and halide perovskite semiconductors. These devices were made on flexible and semitransparent substrates, and exhibited device properties on par with or superior to those obtained on SiO_2 dielectric. The DOS analysis confirmed that the density of trap states at the semiconductor/dielectric interface was slightly increased upon treatment, but it remained lower than the case of SiO_2 interface with the same semiconductor. The method proposed here promotes the incorporation of organic and perovskite semiconductors in all possible configurations of the field-effect transistors, thus expanding their use while taking full advantage of the low-cost associated with solution processing.

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DOI: 10.1021/acsami.7b03232
ACS Appl. Mater. Interfaces 2017, 9, 18120−18126

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The organic semiconductor work at WFU was supported by the National Science Foundation under Grant No. ECCS-1254757, while the work on the perovskite semiconductors was partially supported by the NSF Grant No. ECCS-1608095 and the Office of Naval Research Grant No. N00014-15-1-2943. Semiconductor synthesis was supported by NSF Grant No. DMR-1627428. The Asylum MFP-3D-Bio AFM was obtained with a grant from the North Carolina Biotechnology Center (Grant No. 2014-IDG-1012).

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